2a

Could you briefly share some of the key problems your team has solved using AI technologies?

2b

Given the following AI quality attribute, how important is it in your domain?
(Please rate on a scale of 1 to 5, where 5 = critically important, 4 = highly important, 3 = moderately important, 2 = minimally important, and 1 = not important.)

1. **Correctness** evaluates how reliably the model performs its intended function and produces the expected outputs.

2b

If you consider the AI quality attribute important in the previous question, could you describe the specific challenges you face in ensuring that attribute, and the strategies you use to address or mitigate those challenges?

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2. **Robustness** refers to the model's ability to produce reliable and benign results when exposed to new, unseen, or noisy data and resist adversarial input perturbations.

2b

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3. **Efficiency** denotes a trained AI model's ability to produce outputs with minimal time, computational power, and energy during inference.

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4. **Fairness** refers to the extent to which a model avoids producing unjustified disparities or discriminatory outcomes across different demographic groups or protected attributes.

2b

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5. **Explainability** refers to the ability to explain a model's decision-making processes. It enables humans to understand the factors that contribute to the model's decisions.

2b

If you consider the AI quality attribute important in the previous question, could you describe the specific challenges you face in ensuring that attribute, and the strategies you use to address or mitigate those challenges?

5. Explainability refers to the ability to explain a model's decision-making processes. It enables humans to understand the factors that contribute to the model's decisions.

2b

Given the following AI quality attribute, how important is it in your domain?
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6. **Privacy** concerns the extent to which personal and sensitive data related to the AI model (e.g., training data) is protected from unauthorized access.

2b

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Given the following AI quality attribute, how important is it in your domain?
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7. **Scalability** measures the model's ability to maintain acceptable performance when its workload or data volume increases.

2b

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Given the following AI quality attribute, how important is it in your domain?
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8. **Deployability** measures the ease with which an AI model can be integrated into existing environments, workflows, or systems while maintaining its performance and functionality.

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9. **Maintainability** refers to the ease with which an AI model can be updated, debugged, and improved over time. We focus on operational concerns like model upgrading and monitoring in this study.

2b

If you consider the AI quality attribute important in the previous question, could you describe the specific challenges you face in ensuring that attribute, and the strategies you use to address or mitigate those challenges?

9. **Maintainability** refers to the ease with which an AI model can be updated, debugged, and improved over time. We focus on operational concerns like model upgrading and monitoring in this study.

2c

Is the AI software development workflow that your team adopted significantly different from the classical software development workflow, for example, Scrum or Waterfall? Can you briefly describe your workflow?

4a

Have you utilized **self-collected data**, **purchased data** from other companies, or **open-source datasets** to create training/testing datasets? What factors influenced your decision-making process in selecting these sources?

What components of the AI model development process are fully automated, and what components require human intervention?

3b

Could you provide examples of how you would determine **which attribute of an AI software** (such as correctness, fairness) **to prioritize** based on the **client's requirements**? How would you **align** the client's needs with these aspects?

4c

Do you have experience that you had **limited data** to work with? How did you and your team ensure that your AI models were still **reliable**?

5c

Usually, the data distribution of the target environment of AI software will drift over time. For example, let's consider an AI model for predicting house prices in a certain area. Over time, factors like the local economy, population growth, or even unforeseen events like natural disasters could change the housing market dynamics. The model that was initially trained on past data may no longer provide accurate predictions because the data distribution it was trained on has drifted from the current reality.

In the context of your recent projects, do you believe that they are or could be affected by data drifting over time? If so, how would you approach this issue to ensure that your AI models remain high quality as the underlying data changes? How frequently data update is needed?

5a

What methods will your team use to **validate** and **clean data** in order to ensure data quality?
Will this process influence the **quality** of the AI project?

5d

If you could envision a tool to assist in addressing data quality issues in your AI projects, what key features or capabilities would you desire? How would this ideal tool facilitate your work and improve the overall quality of your data?

8a

In the projects that you have developed, when do you prefer to develop models **from scratch** and when do you prefer to **fine-tune a pretrained model** to assure project quality? What affects your decision-making process?

Under the context that you decided to utilise a pre-trained model:

- Which **source** do you want to search for a suitable PTM? (e.g. Hugging Face, GitHub, PyTorch Hub etc).
- How to find a suitable **model architecture**?
- How to find a suitable **model size & precision**? The bigger the better?

11b

Are **AI-specific logging libraries**, such as mlflow and wandb, being used in your project to track the quality of AI software? If so, how do you use them?

What is the maximum acceptable **failure rate** that your AI software can have?

7a

What strategies would you usually employ to **select the features** that will be involved in the **model input** in order to obtain a high quality? What will influence your decision-making?

8d

What techniques and tools do you use for hyper-parameter tuning?

What tools or frameworks have been most helpful for developing or deploying your AI models?

If the project deadline is tight, how would you prioritize the evaluation criteria and why?

What are the main differences between checking the quality of AI software before deployment and after deployment?

4b

Do you use any method to generate/expand datasets? How do you ensure the quality of the generated data?

6a

What process would your team usually use to organise the data labelling process in order to obtain good data quality while using a relatively low cost?

Have your team managed to obtain feedback from the end users? To what extent can end users' feedback help to improve the AI project's quality?

12e

In the field of AI products, gaining customer **trust** is paramount to the successful deployment and adoption of the solutions. The users need to trust **not only the accuracy** but also the reliability, fairness, and transparency of the models. Could you share your strategies or specific actions taken to **build and maintain this trust** in your AI products?